Find() vs FirstOrDefault()

Primary Key Retrieval

1. Core idea

Use `Find()` for a direct lookup by the entity's primary key. Use `FirstOrDefault()` for flexible queries that may filter by any property or combine conditions.

Find()	FirstOrDefault()
Designed for primary-key lookup.	Works with any LINQ filter condition.
Checks the current DbContext's change tracker first.	Executes the LINQ query against the database when evaluated.
Can return an already tracked entity without sending SQL.	Does not use Find's primary-key/change-tracker lookup behavior.
Returns null/default if the key is not found.	Returns null/default if no matching row is found.

Mental anchor: By primary key → think Find() first. By email, status, or multiple conditions → use FirstOrDefault() or another suitable LINQ query.

2. Why Find() can be faster

1. **Change Tracker first:** if the entity is already tracked by the same DbContext, `Find()` can return it from memory without a database round trip.

2. **Otherwise:** if the entity is not tracked, `Find()` still queries the database. The first lookup is not guaranteed to be faster in every situation.

3. **Repeated lookup:** a later `Find()` for the same key may use the tracked entity, provided it remains tracked in that context.

3. Short examples

FindByPrimaryKey.cs

```
var student = context.Students.Find(1);
```

FirstOrDefaultById.cs

```
var student = context.Students
    .FirstOrDefault(s => s.StudentId == 1);
```

MultipleConditions.cs

```
var student = context.Students
    .FirstOrDefault(s =>
        s.StudentId == 1 &&
        s.Status == "Active");
```

4. Important details

• `Find()` accepts primary-key values only. For composite keys, pass the key values in the configured primary-key order.

• `Find()` is not a general-purpose replacement for `Where()` or `FirstOrDefault()`.

• `Find()` does not support `ToQueryString()` because it is a special key lookup rather than a standard LINQ query. Runtime logging shows SQL only when a database query is actually sent.

5. Common mistake

Wrong approach: trying to use `Find(1, "Active")` to search by ID and status. `Find()` expects primary-key values, not arbitrary filter conditions.

Better: use a LINQ predicate when you need additional conditions, such as `StudentId` plus `Status`.

Next lesson: Selecting only the columns you need with Select(...).